Fiscal Year:	FY 2025	**Task Last Updated:**	FY 06/05/2025
PI Name:	Weil, Michael Ph.D.		
Project Title:	NSCOR: NASA Specialized Center of Research on Carcinogenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Fort Collins	**State:**	CO
Zip Code:	80521-2807	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
Start Date:	06/01/2015	**End Date:**	11/01/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/1/2023 per NSSC information (Ed., 1/23/23) NOTE: End date changed to 5/31/2023 per JSC Grants Office and NSSC information (Ed., 6/2/22) NOTE: End date changed to 5/31/2022 per NSSC information (Ed., 3/3/2020)		
Key Personnel Changes/Previous PI:	Ed. Note: Per the PI, Dr. Thamm has moved off the grant since completing his role on NSCOR. Dr. Story and Dr. Ullrich have been removed from the CoI list as they now have different roles in the grant (8/10/22). April 2020 report: Dr. Thamm has completed his role on the NSCOR. April 2019 report: Dr. Robert L Ullrich has transitioned from NSCOR Co-Director to Consultant; Dr. Michael D. Story now serves as the Co-Director. Dr. Thomas Borak passed away in 2021 and Dr Alexander Brandl joined the NSCOR as a Physicist in 2021.		
COI Name (Institution):	Borak, Thomas Ph.D. (Colorado State University) Emmett, Mark Ph.D. (University Of Texas, Galveston) Hwang, Tae Hyun Ph.D. (University of Texas Southwestern Medical Center at Dallas) Ray, F. Andrew Ph.D. (Colorado State University) Bacher, Jeff Ph.D. (Promega Corporation) Halberg, Richard Ph.D. (University of Wisconsin, Madison) Raber, Jacob Ph.D. (Oregon Health & Science University) Brandl, Alexander Ph.D. (Colorado State University)		

Grant/Contract No.:	NNX15AK13G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The proposed Carcinogenesis NASA Specialized Center of Research (NSCOR) addresses several key questions for the assessment of radiation risk. The NSCOR consists of four interrelated projects. Project 1 is a biomarker discovery study using integrative “omics” approaches over multiple levels of biological organization and involving multiple species. Biomarkers predictive of the outcomes of HZE (high energy) ion exposures can be used to extrapolate findings in mice to other species, including humans, that are most relevant to NASA’s exploratory missions. The biomarkers are also critical for understanding underlying carcinogenic mechanisms, early disease detection, and subsequent countermeasure development. Project 2 investigates qualitative differences in tumor progression and metastasis between HZE ion- and gamma ray-induced tumors. Project 3 examines the critical question of risk from protracted exposures to high LET (linear energy transfer) radiation at low doses and dose rates. To estimate the carcinogenic effects of these scenarios, we will use chronic exposures to high LET associated neutron radiation as a surrogate for conditions of space-relevant fluence rates and total doses. Project 4 utilizes the resources (irradiated mice and “omics” results) generated in the first three projects to study the neurobehavioral consequences of HZE ion and neutron exposures and whether they are related to tumorigenesis-related outcome measures and predicted by the same or distinct biomarkers. Two addenda were added to the NSCOR in the April 2017 – March 2018 reporting period. The first was designed to determine if a low dose aspirin regimen could be an effective countermeasure to high LET radiation-induced hepatocellular carcinoma. Two exposure conditions are included, and acute exposure to 0.2 Gy 300 MeV 28Si ions and a chronic exposure to 0.4 Gy of 252Cf neutrons. The second was designed to compare fractioned exposures to a simulated GCR (galactic cosmic radiation) beam to an acute exposure for the induction of hepatocellular carcinoma.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Accurately determining the cancer risk from high energy, charged particle radiation exposure is of great importance for designing human spaceflight missions, but it is becoming increasingly important for cancer radiotherapy as well. Radiation oncology appears poised to transition to charged particle radiotherapy in the form of proton therapy and carbon ion therapy. However, one of the risks of treating cancer with charged particle radiation is that the treatment itself can result in a new cancers, known as a second malignant neoplasms (SMN) (commonly used photon radiotherapy also increases SMN risk). The radiotherapy equipment and the patient treatment plans are designed to minimize SMN, but the models to predict risks from various exposures rest on some of the same assumptions about how charged particle radiation causes cancer that are being tested in this NSCOR grant. The results obtained in this program can be used to improve the design of treatment protocols and thus reduce the risks of SMN in radiotherapy patients.
Task Progress:	The main goals in the 2022 – 2023 funding period of the Carcinogenesis NASA Specialized Center of Research (NSCOR) were to (1) replace the 252Cf source in the neutron irradiator/vivarium facility, (2) sequence several radiation-induced murine tumors and analyze the data, and (3) analyze the low dose rate neutron exposure data from Project 3. 1. The 252Cf source was replaced in January of 2023. Since the replacement, we have made the facility available to a number of extramural investigators including Mark Boettger and Ryan Sprenger (Fauna Bio), Jon Steller (University of Colorado, Division of Maternal Fetal Medicine), Alexander Meyers (NASA-Kennedy Space Center), Atul Chander, (NASA-Jet Propulsion Lab), Marta Del Bianco (Italian Space Agency), Dindo Mirco (INAF – Observatorio Astrofisico di Arcetri), Nicoletta La Rocca (Universita Degli Studi di Padova), Laura Zucconi (Universita Tuscia), Sarah Wyatt (University of Ohio), and Ruth Farrell (Cleveland Clinic, OB/GYN Institute). 2. Fifty-five hepatocellular carcinomas and 15 thymic lymphomas arising in unirradiated, gamma ray irradiated and HZE ion irradiated mice were selected for genomic sequencing. DNA was extracted and sequenced. The sequences from the HCC have been analyzed. Among the findings are that (1) there are no differences between the radiation-associated and spontaneous HCC for numbers of SNPs plus indels or for structural variants; (2) COSMIC mutational signatures are the same for radiation-associated and spontaneous HCC, though one unique signature each was noted in the gamma ray and spontaneous samples; (3) gene mutations occurred mainly in the RTK-RAS, Notch, PI3K, and WNT pathways; (4) Ctnnb1, Pyhin1, and Lrp1b were among the genes mutated. The data have been submitted for publication. Sixth-eight acute myeloid leukemia (AML) samples were provided to the NASA Translational Radiation Research and Countermeasures (TRRaC) Project team and Dr. Katherine Hoadley at the UNC School of Medicine for sequencing. The AMLs arose in male and female gamma ray and HZE ion irradiated and unirradiated mice. The sequencing has been completed, and analyses are ongoing. 3. We previously found dose-dependent life shortening in female BALB/c mice irradiated with 252Cf neutrons at a low dose rate (1 mGy/day). Similar life shortening did not occur in male C3H mice. The reason for using different strains in the study was to detect a wide spectrum of radiation-induced tumor types. Consequently, we do not know if the lifespan difference we observe is due to a sex difference or a strain difference. This is a key question that further study is needed to resolve. A second question is what radiation-induced pathology causes the life shortening. Twenty tumor types were identified in the female BALB/c mice, but increasing tumor incidence with increasing dose is not responsible for life shortening (though we have not yet analyzed tumor latency). Endometrial stromal polyps did show a trend for a dose-dependent increase in incidence, but they are unlikely to be a cause of death and would not explain life shortening. Identifying the cause(s) of life shortening due to highly protracted, low dose rate exposures to high LET radiation would seem to be of critical importance for understanding the health risks involved in long-duration space missions and developing countermeasures.
Bibliography Type:	Description: (Last Updated: 04/27/2026)

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